

DISORDERS OF DNA REPAIR

Xeroderma Pigmentosum

Xeroderma pigmentosum (XP) serves as the prototype heritable disease with increased sensitivity to cellular injury. It is an autosomal recessive disorder characterized by extreme sun sensitivity, photophobia, early onset of freckling, and a high incidence of skin cancers on sun-exposed areas. Patients exhibit cellular hypersensitivity to ultraviolet (UV) radiation and certain chemicals, due to defects in DNA repair.

Eight clinical phenotypes of DNA repair diseases have been recognized (e.g., XP, Cockayne syndrome, trichothiodystrophy). Mutations in 12 genes are associated with these phenotypes, with significant phenotypic overlap. Notably, mutations in a single gene can lead to multiple distinct clinical syndromes, and different genes can cause similar phenotypes.

For example, mutations in the **XPD (ERCC2)** gene are linked to: - Xeroderma pigmentosum (XP) - XP with neurologic abnormalities - XP-Cockayne syndrome complex - Trichothiodystrophy (TTD) - XP-trichothiodystrophy - COFS syndrome (cerebro-oculo-facio-skeletal syndrome)

The official gene names are listed in parentheses; complementation group names are in uppercase.

Nucleotide Excision Repair and Related Disorders

The **nucleotide excision repair (NER)** pathway repairs bulky DNA lesions, such as UV-induced photoproducts. Defects in NER underlie XP, Cockayne syndrome (CS), and trichothiodystrophy (TTD). Cells from patients with these disorders show increased sensitivity to UV light, with elevated cell death and mutation rates. However, only XP patients have a markedly increased risk of skin cancer. The basis for this differential cancer risk remains unclear.

Some proteins involved in these disorders participate not only in DNA repair but also in transcription and DNA damage response. This dual functionality

explains the overlapping clinical features among XP, CS, and TTD, as mutations may affect different functional domains of the same protein.

XP Variant

The **XP variant** phenotype is clinically indistinguishable from classical XP but results from a defect in **DNA damage tolerance** rather than NER. Specifically, it involves impaired **translesional DNA synthesis**, the process by which normal cells bypass DNA lesions during replication. This defect is typically due to mutations in the **POLH (XPV)** gene, encoding DNA polymerase η .

Mismatch Repair and Cancer Syndromes

The **DNA mismatch repair (MMR)** pathway is defective in **hereditary nonpolyposis colorectal cancer (HNPCC)**, also known as **Lynch syndrome**, and its variant **Muir-Torre syndrome**, which includes cutaneous manifestations such as sebaceous tumors.

Helicases in DNA Repair and Disease

Helicases unwind DNA and are essential for DNA replication, transcription, and repair. Two NER helicases, **XPB (ERCC3)** and **XPD (ERCC2)**, are components of the transcription factor **TFIIH**. XPB unwinds DNA in the 3' to 5' direction, while XPD unwinds in the 5' to 3' direction.

The **RecQ family** of DNA helicases is evolutionarily conserved. Among the five human RecQ helicases, three are associated with genetic disorders: - **BLM** – mutated in Bloom syndrome (BS) - **WRN** – mutated in Werner syndrome (WS) - **RECQL4** – mutated in Rothmund-Thomson syndrome (RTS)

These syndromes are characterized by chromosomal instability, cancer predisposition, and/or features of premature aging.

BASC Complex and Genome Instability

The **BASC (BRCA1-associated genome surveillance complex)** is a nuclear multi-enzyme complex centered on **BRCA1**. It includes key DNA damage response proteins such as: - **ATM - ATR - MRE11-RAD50-NBS1 complex - BRCA1 - BLM - FANCD2 - MLH1**

These proteins physically interact and coordinate the cellular response to DNA damage. Defects in any of these components can lead to genome instability and are associated with disorders including: - Ataxia telangiectasia - Seckel syndrome - Nijmegen breakage syndrome - Hereditary breast and ovarian cancer - Bloom syndrome (BS) - Fanconi anemia (FA) - Hereditary colon cancer

Bloom syndrome and Fanconi anemia are discussed in detail in this chapter.